

AMPGen — AMP Challenge Submission Report

A two-stage generative pipeline for antimicrobial peptide design

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Backbone: ESM2-35M (facebook/esm2_t12_35M_UR50D) · Compute: Modal, single H100

This report documents the AMPGen *submission*: the deliverables, the data behind them, the generator and predictor that produce and rank the library, and how our internal metrics map to the challenge’s scoring. It is the submission-facing companion to the AMPGen method paper ([reports/AMPGen_Report.pdf](#)), which carries the full training methodology and ablations.

1 Submission summary

AMPGen is a two-stage pipeline. **Stage-1** is an activity-blind generator (a domain-adapted masked-language model over the full AMP corpus, used as a Gibbs mutation proposer) that produces natural-like candidate peptides. **Stage-2** is a dual-endpoint predictor on the same frozen encoder that scores each candidate for antibacterial activity (per-species MIC) and safety (human-RBC HC50), and ranks the library. Deliverables:

- A **50,000-sequence FASTA library** of designed peptides satisfying the challenge constraints (20 standard amino acids, length 8–50, linear, free termini).
- A **ranked top-100 list**, ordered by the predictor’s activity/selectivity score, with every entry $\leq 80\%$ identical to any DBAASP/dbAMP/APD reference (novelty gate).
- The generator and predictor weights, training data manifest, and this report.

2 Challenge task and scoring metrics

Submissions are scored by the organisers’ **seqme** library against a fixed **20-strain panel** (15 Gram-negative, 5 Gram-positive; an 8-strain MDR ESKAPE subset) across **five categories**. The table gives what each category rewards and the underlying quantities.

Table 1: Challenge scoring categories and the quantities they reward. Potency threshold = 16 μM ; MIC ceiling = 64 μM ; HC50 ceiling = 128 μM .

Category	What it rewards	Underlying quantity
Broad-Spectrum	Activity across the whole 20-strain panel	Success Rate over all strains
Gram-Positive	Activity on the 5 Gram+ strains	Success Rate / MIC50 (Gram+)
Gram-Negative	Activity on the 15 Gram– strains	Success Rate / MIC50 (Gram–)
MDR ESKAPE	Activity on the 8-strain drug-resistant subset	Success Rate (MDR subset)
Optimal Selectivity	Potent <i>and</i> non-toxic to human cells	Safety Window = HC50 / MIC50

Function of each scoring quantity.

- **Success Rate** — fraction of panel strains for which a peptide reaches $\text{MIC} \leq 16 \mu\text{M}$ (the potency threshold). The core “does it work” measure; drives the four activity categories.
- **MIC50 / MIC90** — median / 90th-percentile MIC across the relevant strain set; summarise potency depth and breadth.
- **Safety Window (HC50/MIC50)** — how much more concentrated a peptide must be to lyse human red blood cells than to kill bacteria; the therapeutic-index proxy that defines the Optimal-Selectivity category.

3 Data sources and scale

All three data assets are unions of curated public databases, de-duplicated and split leak-free (CD-HIT at 80% identity, whole clusters assigned to train or test so no test sequence is >80% identical to any training sequence).

Table 2: Data assets behind the submission. MIC and HC50 feed the two predictor endpoints; the generator corpus is the Stage-1 pretraining set.

Asset	Sources (union)	Seqs	Meas.	Notes
Generator corpus	AMPSphere T1+T2 (physchem-matched) + DRAMP + DBAASP + dbAMP + APD	155,280	—	all-AMP MLM pretrain
MIC (10 species)	GRAMPA + ampbench-v0.7 master + AMPBench-MT	13,123	40,874	$\log_{10} \mu\text{M}$; 14 flags free-termini
HC50 (human RBC)	ampbench master hc50_human (DBAASP+HemoPI2+Hemolytik+QMAP+DRAMP) + Hemolytik2 + DRAMP text	4,402	4,402	14.2% right-censored

Table 3: MIC measurement scale per panel species (post-union, 10 species). AMPBench-MT contributes the DBAASP rare-species MIC our earlier GRAMPA+master build did not include, multiplying the low-data species 3–13 $\times$.

Species	Gram	measurements	panel strains
<i>E. coli</i>	—	11,143	5
<i>S. aureus</i>	+	9,776	2
<i>P. aeruginosa</i>	—	6,977	3
<i>B. subtilis</i>	+	3,139	1
<i>S. enterica</i>	—	2,393	2
<i>K. pneumoniae</i>	—	2,353	2
<i>A. baumannii</i>	—	2,025	2
<i>E. faecalis</i>	+	1,779	1
<i>E. faecium</i>	+	771	1
<i>E. cloacae</i>	—	518	1
Total		40,874	20

4 Stage-1 — Generator

The generator is a domain-adaptive masked-LM pretrain of ESM2-35M on the 155,280-sequence all-AMP corpus (variable-rate span/scattered masking, $1/t$ -reweighted cross-entropy, 3 epochs), used as a temperature-sampled mutation proposer. It is judged on whether its output is *natural-like* on the axes that matter for AMPs (physicochemical marginals and an independent language-model naturalness score), not on exact-residue recovery.

The generator reproduces the natural AMP marginal (physchem Wasserstein near zero on every axis, pseudo-perplexity matched to natural) and passes the novelty gate at 100%, i.e. it neither collapses to a degenerate mode nor drifts out-of-distribution.

Table 4: Generator (Stage-1) held-out metrics. Reference = held-out natural AMPs; pseudo-perplexity is from an independent ESM2-150M, distinct from the generator backbone.

Metric	init	final	reference / comparator
Validation loss	2.633	2.236	—
Exact masked-token recovery	0.193	0.277	chance 0.05 ($> 3\times$)
Novelty, fraction $\leq 80\%$ identity	—	100%	challenge novelty gate
Embedding diversity (mean pairwise cos-dist)	—	0.0977	natural 0.1009 (matched)
Pseudo-perplexity, median	—	17.12	natural 17.01 (near-identical)
Wasserstein: len / charge / GRAVY / μH / hydrophobic	—	0.000 / 0.099 / 0.014 / 0.005 / 0.002	(lower = closer)

5 Stage-2 — Dual-endpoint predictor

On the frozen Stage-1 encoder we train LoRA ($r=8$, $\alpha=16$, last 4 attention layers; 0.79 M trainable params, 2.3%) feeding two heads: a **species-conditioned MIC head** (AdaLN over 10 species) predicting $-\log_{10}\text{MIC}$, and a **single HC50 head** predicting $\log_{10}\text{HC50}$. Both are trained to *absolute* $\log_{10} \mu\text{M}$ with a composite loss (pairwise ranking + Huber magnitude regression); the HC50 loss is censoring-aware (right-censored “safe-tail” measurements penalised one-sidedly). The two heads are trained as **separate** models (shared frozen encoder, own LoRA) — multitask training gave no MIC gain and a clear HC50 loss (the small-task seesaw), so separate is the final configuration.

5.1 Version progression (v1 $\rightarrow$ v3)

Table 5: Predictor performance across iterations, overall Spearman ρ / Pearson r on the honest (homology) split, plus the random-split ρ . v1 = dual-endpoint baseline; v2 adds rare-species sampling rebalance and censoring-aware HC50 loss; v3 adds the AMPBench-MT data merge.

Ver	Change	MIC ρ/r (hon)	HC50 ρ/r (hon)	MIC ρ (rand)	HC50 ρ (rand)
v1	dual-endpoint baseline	0.578 / 0.592	0.456 / 0.478	0.645	0.597
v2	+ rebalance + censoring	0.585 / 0.597	0.503 / 0.522	0.637	0.563
v3	+ AMPBench-MT data	0.494 / 0.512 [†]	0.421 / 0.459 [†]	0.619	0.592

[†] **v3 honest numbers are confounded by re-clustering.** Adding 16,821 MIC rows changed the global CD-HIT clustering, so the v3 honest test set differs from v2’s and is harder. The proof: HC50’s data is *unchanged* between v2 and v3, yet its honest ρ moved $0.503 \rightarrow 0.421$ (-0.082) purely from the re-split. MIC’s -0.091 is therefore almost entirely the harder split, not a regression. Net: the data merge is ρ -neutral on the overall honest split while tripling-to-13 $\times$ -ing rare-species coverage — a coverage gain for the MDR/Gram categories rather than a headline- ρ gain. **v2 remains the recommended configuration by honest overall ρ** ; the v3 data is retained for its rare-species coverage.

5.2 Per-species MIC (honest split)

6 Our evaluation metrics and their function

The predictor is scored with four complementary metrics, on two splits, against a mean-predictor baseline — so that a number can never be read as good without its honest reference.

- **Spearman ρ** (ranking, calibration-free) — how well the score *orders* peptides by activity/safety. This is what the **top-100 ranking** deliverable depends on.

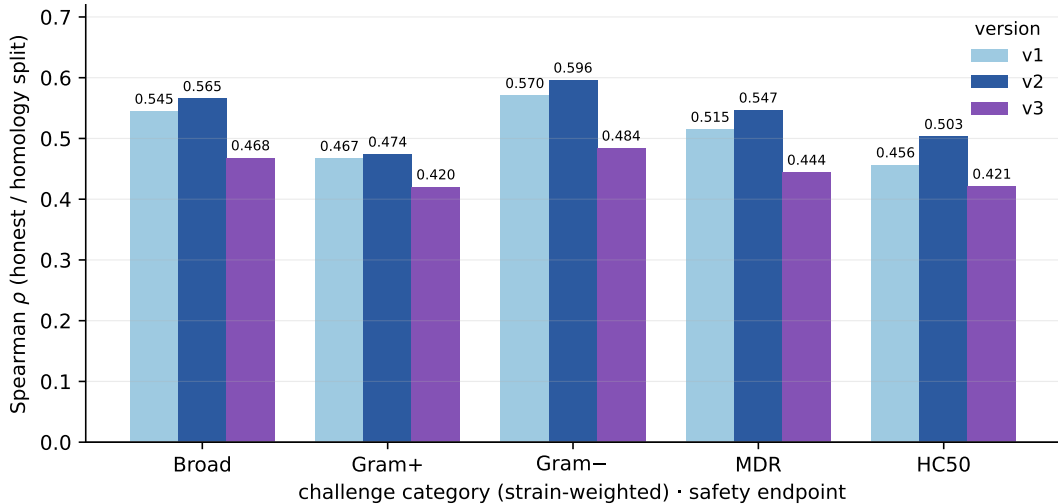


Figure 1: Predictor performance across versions v1–v3 (bar colour) on the honest homology split, at the *challenge-category* level: the four MIC categories are strain-count-weighted means of the per-species Spearman ρ (via the panel aggregation of §7), and HC50 is the safety endpoint. v2 is best on every category; v3 dips because its re-clustered test set is harder (see †), not because the added data hurts.

Table 6: MIC Spearman ρ per panel species on the honest split for the recommended v2 model, with the v3 test-set size showing the coverage growth from the AMPBench-MT merge. Low- n species carry wide confidence intervals.

Species	v2 ρ	v2 test- n	v3 test- n
<i>E. coli</i>	0.598	937	1142
<i>S. aureus</i>	0.550	774	1036
<i>P. aeruginosa</i>	0.631	620	641
<i>B. subtilis</i>	0.546	90	338
<i>S. enterica</i>	0.578	48	231
<i>K. pneumoniae</i>	0.590	47	207
<i>E. faecalis</i>	0.255	39	165
<i>E. cloacae</i>	0.356	25	68
<i>E. faecium</i>	0.469	19	80
<i>A. baumannii</i>	0.681	13	157

- **Pearson r** (magnitude, on absolute $\log_{10}\mu\text{M}$) — whether predicted values track true values in *absolute* terms. Required because the challenge scores an absolute 16 μM potency threshold and the HC50/MIC50 safety window; a rank-only model cannot supply these. We measure r on *raw* values, not by inverting a ranking score. Because $\rho \approx r$ (not $\rho \approx 2r$), the model genuinely calibrates magnitude.
- **RMSE / MAE** ($\log_{10}\mu\text{M}$) — absolute error, reported against the mean-predictor’s RMSE (its ceiling when $\rho=r=0$).
- **Split.** The *homology* split (CD-HIT-80% whole-cluster) is the honest, deployment-relevant number; the *random* by-sequence split leaks homologs across train/test and is the inflated, literature-comparable number. We always report both.

7 Panel aggregation — species to category scores

The predictor emits one MIC per species and one HC50; the challenge scores a 20-strain panel. The aggregation (`challenge/metrics/aggregate.py`) broadcasts each species’ predicted MIC to the

panel strains it represents (PANEL_STRAIN_COUNTS: *E. coli*×5, *P. aeruginosa*×3, ...) and computes, per category, **Success Rate** (fraction of strains with MIC $\leq 16 \mu\text{M}$), **MIC50/MIC90** (strain-weighted percentiles), and, for Optimal-Selectivity, **Safety Window** = HC50 / MIC50 — all with the challenge ceilings (MIC 64 μM , HC50 128 μM) applied. Categories: Broad (all 20 strains), Gram+ (5), Gram− (15), MDR (the 8-strain ESKAPE subset). The same category weighting yields the category-level ranking ρ shown in Figure 1 and Table 6, where v2 leads every category.

Table 7: Category-level ranking quality: strain-count-weighted mean of per-species MIC Spearman ρ per challenge category (honest homology split), plus the HC50 safety endpoint, for each version. v2 is best on every category; v3’s dip is the re-clustering artifact († above).

Version	Broad	Gram+	Gram−	MDR	HC50
v1	0.545	0.467	0.570	0.515	0.456
v2	0.565	0.474	0.596	0.547	0.503
v3	0.468	0.420	0.484	0.444	0.421

8 Decision and status

Final predictor: v2 configuration (separate mode) — honest MIC ρ 0.585 / r 0.597 (RMSE 0.65 $\log_{10}\mu\text{M}$), honest HC50 ρ 0.503 / r 0.522 — with the v3 AMPBench-MT data retained for its 3–13× rare-species coverage (valuable for the MDR ESKAPE and Gram categories). The censoring-aware HC50 loss was a clear win (ρ +0.047); the rare-species sampling rebalance and the data merge together lift the data-starved species that the challenge’s resistance-focused categories weigh most. A physicochemical-feature branch was diagnosed out (incremental r +0.015 MIC / ~ 0 HC50 — ESM2 already subsumes the descriptors).

Remaining work toward submission: the fixed-seed 50,000 generation run (constraints as a generation gate), the top-100 selection under the novelty gate (ranked by the aggregated selectivity score of §7), and a **seqme** scoring mirror for local validation.